

TRONDHEIM VAERNES, Norway

WMO: 012710

Lat: 63.459N

Lon: 10.935E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-13.5	-19.9	0.6	-15.2	-17.3	0.8	-11.8	16.4	2.1	13.9	2.0	3.5	120	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	25.2	16.8	23.2	16.3	21.2	15.3	18.2	23.0	17.1	21.7	16.0	20.2	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.5	20.5	15.1	10.7	19.3	14.0	10.0	18.0	51.3	23.1	48.0	21.8	44.9	20.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.6	9.1	-18.4	28.3	3.2	2.2	-20.6	29.9	-22.5	31.2	-24.3	32.4	-26.6	34.1		
(5)			-18.7	19.7	3.1	1.5	-20.9	20.8	-22.7	21.6	-24.5	22.4	-26.7	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-1.5	-1.4	0.8	5.2	9.2	12.4	15.4	14.7	11.3	6.1	1.5	-1.0
	DBStd	7.35	5.23	4.99	4.06	3.39	3.74	3.19	3.27	3.11	3.18	3.68	4.67	5.37
	HDD10.0	1930	357	318	286	150	62	10	1	1	22	129	256	339
	HDD18.3	4490	616	551	544	395	285	179	105	120	213	379	505	598
	CDD10.0	505	0	0	0	5	36	82	167	147	59	8	1	0
	CDD18.3	24	0	0	0	0	1	2	13	8	1	0	0	0
	CDH23.3	168	0	0	0	0	6	16	94	51	1	0	0	0
CDH26.7	27	0	0	0	0	0	1	19	7	0	0	0	0	
Wind	WSAvg	3.9	4.3	4.2	4.2	3.8	3.9	4.0	3.6	3.5	3.6	3.7	3.8	4.0
Precipitation	PrecAvg	1012	76	77	78	58	63	95	100	97	105	91	81	91
	PrecMax	1225	199	163	189	132	96	148	175	168	243	172	226	192
	PrecMin	665	0	7	27	21	33	35	47	41	39	35	11	19
	PrecStd	133	45	38	41	27	19	27	36	35	55	30	58	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	8.8	10.6	18.5	24.1	25.5	29.3	27.4	22.3	17.9	12.9	9.7
		MCWB	4.4	5.2	5.7	10.4	14.0	15.6	18.9	17.1	14.7	11.9	9.0	6.1
	2%	DB	6.9	7.2	8.6	15.0	20.8	22.8	25.8	24.7	20.0	15.1	10.6	8.0
		MCWB	4.0	4.1	4.9	8.6	13.2	15.4	17.6	17.1	13.9	10.3	7.3	4.9
	5%	DB	5.9	6.1	7.2	12.7	18.0	20.4	23.6	22.3	18.0	13.4	9.0	6.9
		MCWB	3.2	3.4	4.1	7.6	11.9	14.2	17.0	16.3	13.0	9.4	5.9	4.1
	10%	DB	4.8	4.9	6.1	10.8	15.8	18.2	21.4	20.3	16.3	11.5	7.8	5.7
		MCWB	2.6	2.7	3.4	6.5	10.4	13.3	16.1	15.3	12.2	8.3	5.1	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.0	6.7	10.9	15.5	17.5	20.4	19.0	16.2	12.6	9.5	7.0
		MCDB	6.7	7.5	8.9	16.9	21.0	22.9	26.9	24.3	20.5	16.6	11.6	8.6
	2%	WB	4.6	4.8	5.7	9.2	13.8	15.9	18.7	17.8	14.7	10.9	7.9	5.5
		MCDB	6.1	6.5	7.7	13.8	19.6	21.2	23.9	22.5	18.4	14.3	10.2	7.2
	5%	WB	3.6	3.7	4.6	8.1	12.3	14.8	17.6	16.8	13.7	9.8	6.3	4.5
		MCDB	5.3	5.5	6.8	11.6	17.5	19.3	22.2	21.3	16.9	12.8	8.3	6.2
	10%	WB	2.7	2.8	3.7	6.9	10.8	13.7	16.5	15.8	12.7	8.7	5.2	3.5
		MCDB	4.3	4.5	5.6	10.3	15.1	17.8	20.7	19.6	15.7	11.3	7.4	5.2
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.6	6.5	7.4	7.9	7.3	7.8	7.7	7.1	6.1	5.2	5.5
		MCDBR	5.3	5.6	6.8	10.5	11.7	11.7	11.9	11.3	9.7	8.0	6.5	5.7
		MCWBR	3.9	3.9	4.6	5.9	6.1	6.1	5.7	5.7	5.2	4.9	4.6	4.3
	5% WB	MCDBR	5.1	5.2	6.3	9.2	11.1	10.2	11.1	10.1	8.5	7.1	6.2	5.4
		MCWBR	4.1	4.2	4.6	5.6	6.1	5.7	5.8	5.6	5.2	4.8	4.8	4.4
Clear-Sky Solar Irradiance	taub	0.229	0.290	0.322	0.350	0.354	0.354	0.363	0.350	0.329	0.307	0.366	0.357	
	taud	2.035	2.210	2.251	2.341	2.418	2.457	2.454	2.484	2.531	2.436	2.684	2.611	
	Ebn at Noon	376	635	767	829	860	867	847	826	775	631	359	201	
	Edh at Noon	29	60	86	98	100	98	96	85	67	49	26	17	
All-Sky Solar Radiation	RadAvg	0.15	0.68	1.93	3.72	4.56	4.75	4.62	3.55	2.13	0.94	0.25	0.07	
	RadStd	0.02	0.10	0.28	0.30	0.31	0.49	0.48	0.42	0.22	0.13	0.04	0.01	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page